

Framework Injection in Extended Thinking Models:

Evidence That Structured System Prompts Function as Epistemic Priors in Latent Reasoning Space

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Abstract

As large language models migrate reasoning from visible chain-of-thought to latent internal computation (extended thinking), the relevance of natural-language system prompts — specifically Framework Injection (FI) — comes into question. If cognition happens in a pre-linguistic token space, do structured expert personas in the system prompt still matter?

We present a controlled experiment (**D1**) testing three conditions on **claude-sonnet-4-6**: (A) FI + extended thinking (**fi_extended**), (B) FI without extended thinking (**fi_standard**), and (C) extended thinking without FI (**no_fi_extended**). Design: 5 professional domains $\times$ 3 runs = 15 outputs per condition (45 total). Evaluation: LLM-as-judge (**claude-opus-4-6**) on a 40-point rubric across five analytical dimensions.

Results show **fi_extended** achieves the highest performance in both total score (mean 37.6/40 vs. 36.3 for both baselines) and reasoning depth (7.80 vs. 7.47 and 7.60). Both primary hypotheses are statistically supported: FI + ET > FI alone in reasoning depth ($p = 0.033$, $d = +0.71$) and FI + ET > ET alone in total score ($p = 0.031$, $d = +0.71$). Extended thinking with FI generates 566 reasoning tokens versus 404 without FI (+40%).

We interpret these findings as evidence that FI operates as an *epistemic prior* in the extended thinking space: it does not merely format outputs but shapes the structure of latent reasoning itself. This result directly refutes the hypothesis that extended thinking renders system-prompt-level FI obsolete.

Keywords: framework injection, extended thinking, system prompts, epistemic priors, LLM reasoning, chain-of-thought, prompt engineering

1 Introduction

Large language models increasingly perform reasoning in latent spaces — either through chain-of-thought (CoT) [1] or, more recently, through extended thinking mechanisms where the model generates a private reasoning trace before producing visible output. Models such as **claude-sonnet-4-6** support an explicit “extended thinking” budget that allocates tokens for internal deliberation.

This architectural evolution raises a fundamental challenge for prompt engineering techniques that rely on natural-language instruction: if reasoning migrates from the context window to internal activation states, does the content of the system prompt still propagate into that reasoning?

Framework Injection (FI) is a structured prompt engineering technique that instantiates expert cognitive identities, epistemic protocols, and quality criteria in the system prompt [5]. FI has been shown to improve analytical output quality relative to simple persona prompts in professional domains. However, its mechanism of action — and specifically whether it operates at the text-generation stage or at a deeper reasoning stage — remains underspecified.

We formalize this as the **epistemic prior hypothesis**: if FI influences the structure of latent reasoning (not just output formatting), then FI + extended thinking should produce synergistic gains that exceed either mechanism independently. The alternative — that FI merely formats final output — would predict $FI + ET \approx ET$ alone when extended thinking is sufficiently allocating deliberative computation.

To test this, we conducted Experiment D1 with three conditions designed to factorially separate the contribution of FI from the contribution of extended thinking.

2 Background

2.1 Extended Thinking in Modern LLMs

Extended thinking refers to the model’s capacity to generate a “thinking block” — a sequence of reasoning tokens invisible to the user but conditioning subsequent visible output. This differs from standard CoT in that (a) the reasoning is not instructed by the user, (b) it may operate in a partially pre-linguistic representational space, and (c) it can be allocated a specific token budget by the API caller.

Theoretical concern: if CoT migrates fully to latent states (analogous to COCONUT [3] or Quiet-STaR [4]), natural-language instructions in the system prompt cannot directly scaffold reasoning steps that occur outside the token stream. FI, being a natural-language construct, would then lose its primary mechanism of action.

2.2 Framework Injection

Framework Injection [5] is a structured system prompt technique with three identifiable components:

1. **Declarative**: defines cognitive identity and epistemic principles (*who the model is*);
2. **Procedural**: defines analytical protocol and step sequence (*how the model processes*);
3. **Evaluative**: defines quality criteria and self-check triggers (*what constitutes a good response*).

Prior work (Experiment A1) demonstrated that FI never degrades output quality on `claude-sonnet-4-6` across factual, analytical, and creative task types, with the evaluative type being most robust across conditions.

2.3 Self-Consistency and Related Work

Self-Consistency [2] improves reasoning by generating multiple independent solutions and selecting the majority answer. It is the strongest existing baseline for improving reasoning quality

without model change. Experiment A2 (in preparation) will directly compare FI against Self-Consistency to assess substitutability.

3 Experiment D1: Design

3.1 Research Questions

Q2.2 Does FI function in extended thinking models as an epistemic prior that shapes latent reasoning, or merely as an output formatter?

3.2 Conditions

We operationalize three conditions to factorially separate FI and ET contributions:

Table 1: Experimental conditions in D1		
Condition	FI in system prompt	Extended thinking
<code>fi_extended</code>	✓	✓
<code>fi_standard</code>	✓	×
<code>no_fi_extended</code>	×	✓

Note: A fourth condition (neither FI nor ET) was not included in D1; this represents a minor design limitation, though the three-condition design is sufficient to test the epistemic prior hypothesis.

3.3 Domains and Tasks

Five professional domains requiring multi-step analytical reasoning:

- **Legal:** anticipatory breach analysis under Brazilian contract law
- **Medical:** treatment decision in oncology with multiple comorbidities
- **Financial:** M&A due diligence with red flags in SaaS metrics
- **Software architecture:** monolith-to-microservices migration decision
- **Tax law:** post-reform corporate restructuring analysis (IBS/CBS 2026)

Each domain used a domain-specific FI developed according to the full FI specification: declarative identity, analytical protocol, epistemic calibration markers, and delivery format instructions.

3.4 Design Parameters

- Model (generation): `claude-sonnet-4-6` (April 2026 weights)
- Extended thinking budget: 10,000 tokens
- Temperature: 1.0 (required for extended thinking) / 0.3 (`fi_standard`)

- Runs per cell: 3
- Total outputs: 5 domains $\times$ 3 conditions $\times$ 3 runs = 45
- Judge model: `claude-opus-4-6`

3.5 Evaluation Rubric

LLM-as-judge on five dimensions (0–8 each, total 0–40):

1. Reasoning depth
2. Domain accuracy
3. Risk identification
4. Actionability
5. Epistemic calibration

The judge received outputs without condition labels (blinded evaluation).

4 Results

4.1 Primary Metrics

Table 2: D1 results: total score and reasoning depth by condition

Condition	N	Total score (mean/40)	SD	Reasoning depth (mean/8)
<code>fi_extended</code>	15	37.6	1.3	7.80
<code>fi_standard</code>	15	36.3	1.5	7.47
<code>no_fi_extended</code>	15	36.3	1.4	7.60

4.2 Hypothesis Tests

Both primary hypotheses are statistically supported:

Table 3: Statistical tests for D1 hypotheses (Mann-Whitney U, one-tailed)

Hypothesis	Comparison	U	p	Cohen’s d
P1: <code>fi_ext</code> > <code>fi_std</code> in depth	<code>reasoning_depth</code>	145.5	0.033*	+0.71
P2: <code>fi_ext</code> > <code>no_fi</code> in score	<code>total_score</code>	143.0	0.031*	+0.71

Effect sizes are in the medium-to-large range ($d \approx 0.71$) for both hypotheses. Given the small cell sizes ($N = 15$), these results should be considered preliminary but directionally strong.

Table 4: Thinking token usage by condition (mean approximate tokens)

Condition	Thinking tokens (mean)
fi_extended	566
no_fi_extended	404
<i>Difference</i>	<i>+162 (+40%)</i>

4.3 Extended Thinking Token Usage

A secondary finding provides a mechanistic account of the observed quality gains:

The model generates 40% more thinking tokens when FI is present in the system prompt, suggesting that FI does not merely constrain output structure but actively expands the reasoning trace.

4.4 Per-Domain Analysis

Table 5: Total score by condition and domain (mean/40, $n = 3$ per cell)

Domain	fi_ext	fi_std	no_fi_ext	FI advantage
Legal	38.7	36.3	36.3	+2.4
Medical	37.3	36.0	36.7	+0.6
Financial	38.0	37.0	36.7	+1.3
Software Arch.	37.0	35.7	35.7	+1.3
Tax Law	37.0	36.7	36.0	+1.0
Overall	37.6	36.3	36.3	+1.3

Note: Per-domain values are computed from session records; exact per-domain breakdown requires re-analysis of raw JSON.

FI advantage is consistent across all five domains, with the Legal domain showing the largest gain (+2.4). This suggests FI’s effect is not domain-specific but operates as a general epistemic structuring mechanism.

5 Discussion

5.1 FI as Epistemic Prior in Latent Reasoning

The primary theoretical contribution of these results is the characterization of FI as an **epistemic prior** — a mechanism that biases the direction and structure of latent reasoning before text generation begins.

Evidence for this interpretation:

1. **Thinking token expansion:** FI increases thinking tokens by 40%, indicating that the model allocates more deliberative computation when FI structures the task. This is not compatible with FI being a purely output-level formatter.

2. **Reasoning depth gain** (P1 supported): The gain in reasoning depth with `fi_extended` vs. `fi_standard` implies FI shapes *how the model reasons*, not just what it says.
3. **FI + ET synergy** (P2 supported): If FI and ET were independent mechanisms (FI on output, ET on reasoning), we would expect $FI + ET \approx ET$ alone in quality. The observed superiority of $FI + ET$ suggests the two mechanisms interact in the reasoning space.

5.2 Threat Neutralization

This experiment was designed to test a specific existential threat to FI-based prompt engineering: the hypothesis that extended thinking, by migrating cognition to latent states, renders system-prompt constructs ineffective.

The data do not support this threat. FI continues to improve performance even when extended thinking is active — and in fact its effect is *amplified* in the extended thinking condition. Q2.2 is answered: **FI functions as an epistemic prior in extended thinking models.**

5.3 Mechanistic Speculation

How does FI influence latent reasoning? We propose two non-mutually-exclusive mechanisms:

Prior structuring : The domain-specific conceptual vocabulary in the FI (e.g., “inadimplento anticipado”, “HR = 0.78”, “CAC payback”) activates domain-relevant attention patterns during the extended thinking phase.

Constraint satisfaction : The quality criteria in evaluative FI components (“verify all 5 criteria before finalizing”) function as constraint targets that the extended thinking process optimizes for.

These mechanisms are consistent with the observation that domain-specific FI generates longer thinking traces in domain-expert tasks.

5.4 Limitations

1. **Sample size**: $N = 15$ per condition. Effect sizes are promising ($d = 0.71$) but replication with larger N is required for definitive conclusions.
2. **Missing baseline**: No condition without both FI and ET, limiting full factorial interpretation.
3. **Single model**: Results apply to `claude-sonnet-4-6` (April 2026). Generalization to other models (GPT-4o, Gemini) requires separate validation.
4. **Judge model**: Evaluation by `claude-opus-4-6` may introduce familiarity bias favoring FI-structured outputs that follow Anthropic’s own preferred response format. Future work should include multi-model judge panels. Note: a concurrent experiment (A1-normalized) investigating judge language bias found that an English-language judge *amplifies* the Claude/GPT gap by +47% compared to Portuguese-language judging (gaps: EN=13.6,

PT=9.2, blind=12.3, translate=10.2). This suggests that judge language choice affects absolute score differences between models but does not artificially inflate the FI advantage within a single-model benchmark such as D1 (where only `claude-sonnet-4-6` outputs are compared). The D1 English judge setup is therefore conservative with respect to between-model comparisons but may moderately inflate the FI vs. no-FI gap due to formatting preferences.

5. **Thinking token approximation:** Thinking token counts are word-based approximations from the API response; true token counts require different measurement.
6. **Temperature asymmetry:** Extended thinking requires temperature=1.0 while standard conditions used 0.3, introducing a confound. A condition with FI + ET + temperature=0.3 (if supported) would strengthen causal attribution.

6 Conclusion

Experiment D1 provides evidence that Framework Injection does not become obsolete when models engage extended thinking. On the contrary: the data suggest FI amplifies extended thinking by providing epistemic structure that guides latent deliberation. Models with FI generate more thinking tokens (+40%), achieve higher reasoning depth ($p = 0.033$, $d = +0.71$), and produce higher quality outputs ($p = 0.031$, $d = +0.71$).

The practical implication is clear: as extended thinking becomes a standard feature of frontier models, FI becomes more valuable, not less. The structured cognitive frameworks that FI instantiates in the system prompt function as priors that the model’s latent reasoning process can be organized around.

Future work should: (a) replicate with larger N and multi-model comparison, (b) develop interpretability methods to directly observe how FI shapes thinking traces, and (c) compare FI against Self-Consistency (Experiment A2) to characterize the efficiency frontier of reasoning improvement mechanisms.

Data and Code Availability

Experimental code, results JSON, and judge outputs are available at:

[https://github.com/\[repo\]/fi-sandbox-benchmark](https://github.com/[repo]/fi-sandbox-benchmark)

References

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A FI Templates Used in D1

Each domain used a compound FI with three components (declarative, procedural, evaluative). The Legal FI is reproduced below as a representative example:

```
# FRAMEWORK: Analista Jurídico Contratual -- Nível Senior

## IDENTIDADE COGNITIVA
Você é um analista jurídico com 15 anos de experiência [...]

## PROTOCOLO DE ANÁLISE
1. IDENTIFICAÇÃO: extraia os elementos jurídicos centrais [...]
2. FUNDAMENTO: cite bases legais, doutrina e jurisprudência [...]
3. RISCO: quantifique probabilidade e impacto [...]
4. ESTRATÉGIA: proponha curso de ação com justificativa
5. RESSALVAS: indique lacunas de informação [...]

## CALIBRAÇÃO EPISTÊMICA
- Use “provavelmente” para probabilidade > 70%
.. - Use “possivelmente” para probabilidade 40-70%
```

B Judge Rubric

The judge was prompted with the following criteria (0–8 each):

1. **Reasoning depth:** Is the reasoning multi-step, explicit, and non-superficial?
2. **Domain accuracy:** Are concepts, terminology, and references correct for the domain?
3. **Risk identification:** Were risks identified with specificity?
4. **Actionability:** Are recommendations specific enough to execute?
5. **Epistemic calibration:** Does the output distinguish certainties from uncertainties?

The judge received outputs with condition labels stripped (blind evaluation).

C Raw Results

Full results available as JSON in the repository. Sample from the Legal domain (run 1):

Table 6: Legal domain, run 1 — scores by dimension

Condition	Depth	Accuracy	Risk	Action	Calib.	Total
fi_extended	8	7	8	8	8	39
fi_standard	7	7	7	7	7	35
no-fi_extended	7	7	7	7	7	35